

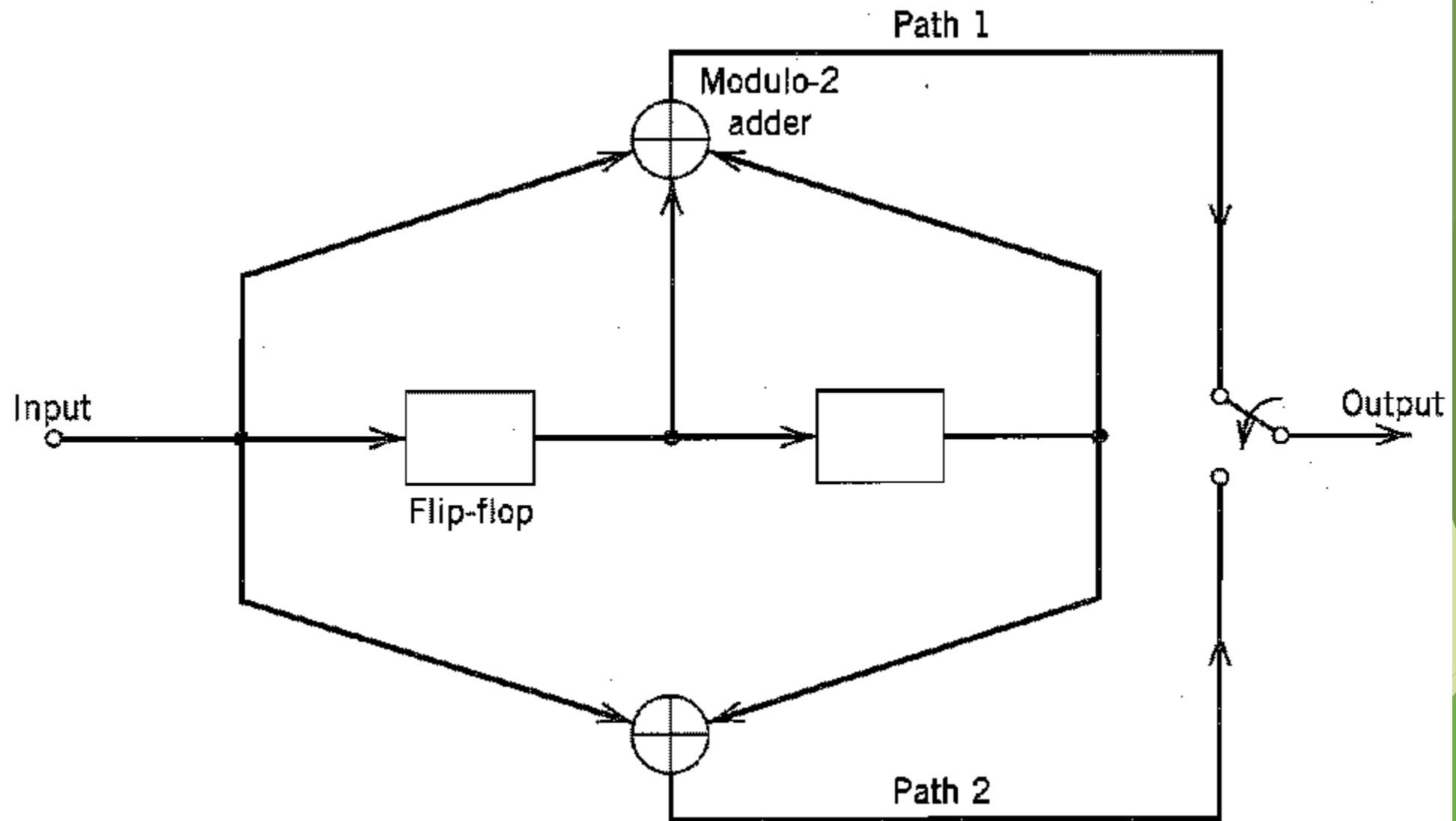
Convolutional Codes

Introduction

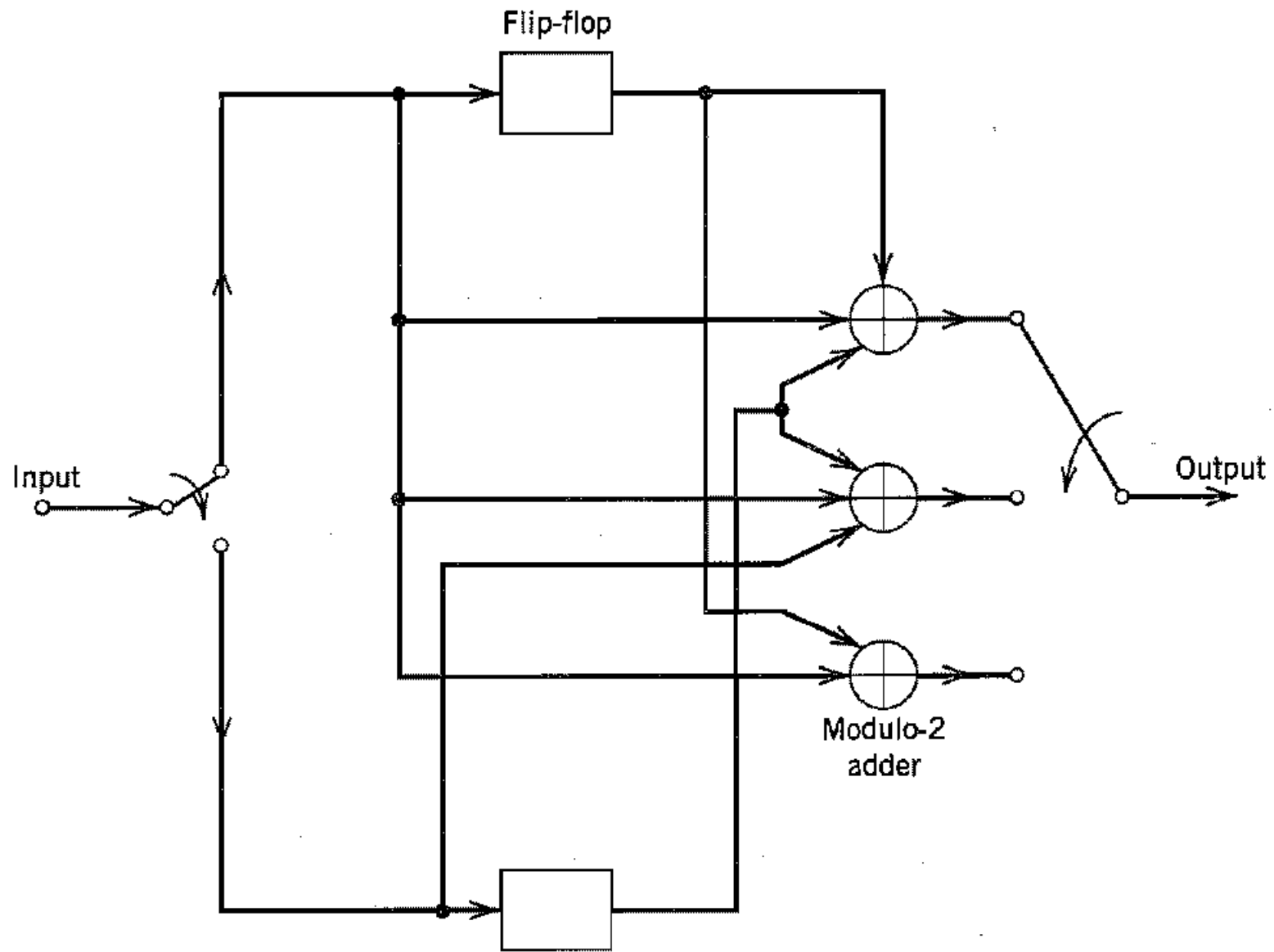
- ▶ In an (n, k) LBC, the encoder buffer an entire message block (of length k) before generating an associated codeword (of length n).
- ▶ In convolutional encoder, the message bits come in serially rather than in large blocks and it's a memory encoder.
- ▶ The encoder of a binary convolutional encoder may be viewed as a *finite-state-machine* that consists of M -stage shift register with prescribed connections to n modulo-2 adders and a multiplexer that serializes the output of the ladder.
- ▶ An L -bit message sequence produces a coded output of length $n(L + M)$ bits. The code rate is therefore

$$r = \frac{L}{n(L+M)} \text{ bits/symbol}$$

- ▶ Typically $L \gg M$. Hence the code rate is $r \approx \frac{1}{n}$ bits/symbol.
- ▶ The constraint length ($K = M + 1$) of the encoder is the number of shifts over which a single message bit can influence the encoder output.



Constraint length-3, rate- $\frac{1}{2}$ encoder

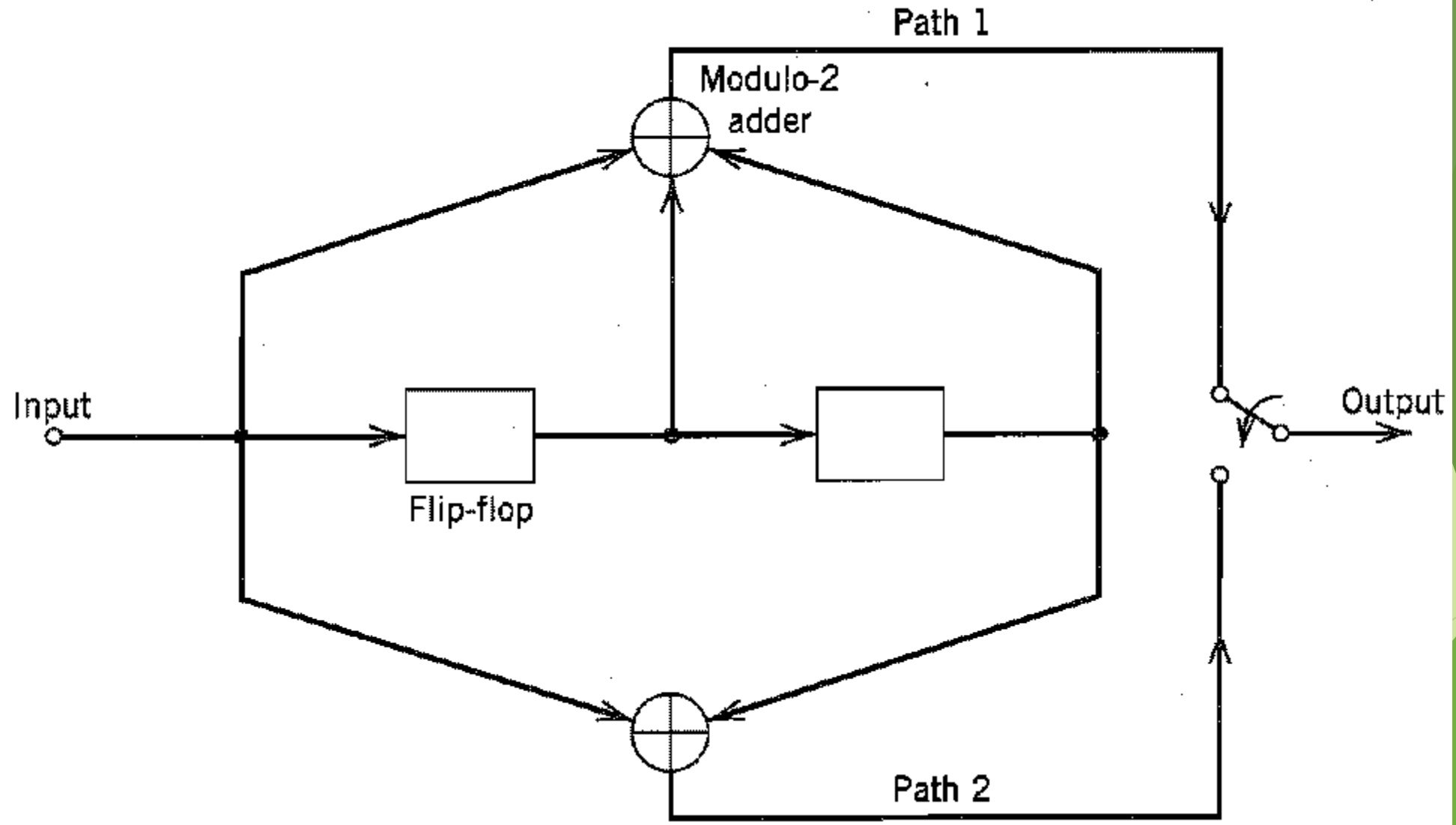


Constraint length-2, rate- $\frac{2}{3}$ encoder

- ▶ The response of a path to a symbol 1 applied to its input, with its each flip-flop set initially in the zero state, is called impulse response.
- ▶ The generator polynomial characterize the unit-delay transform of the impulse response.
- ▶ Letting $g_i^{(j)} = (g_{i,0}^{(j)}, g_{i,1}^{(j)}, \dots, g_{i,m}^{(j)})$ represent the generator sequences corresponding to i^{th} input and j^{th} output.
- ▶ The generator polynomial of the j^{th} path from i^{th} input is
$$g_i^{(j)}(D) = g_{i,0}^{(j)} + g_{i,1}^{(j)}D + g_{i,2}^{(j)}D^2 + \dots + g_{i,M}^{(j)}D^M$$

Where D denotes unit delay variable.

Example



Constraint length-3, rate- $\frac{1}{2}$ encoder

- ▶ The impulse response of path 1 (i.e., upper path) is $(1,1,1)$. The corresponding generator polynomial is

$$g^{(1)}(D) = 1 + D + D^2$$

- ▶ The impulse response of path 2 (i.e., lower path) is $(1,0,1)$. The corresponding generator polynomial is

$$g^{(2)}(D) = 1 + D^2$$

- ▶ For the message sequence (10011) , say, we have the polynomial representation $m(D) = 1 + D^3 + D^4$

- ▶ The output of path 1 is given by

$$c^{(1)}(D) = g^{(1)}(D)m(D) = 1 + D + D^2 + D^3 + D^6$$

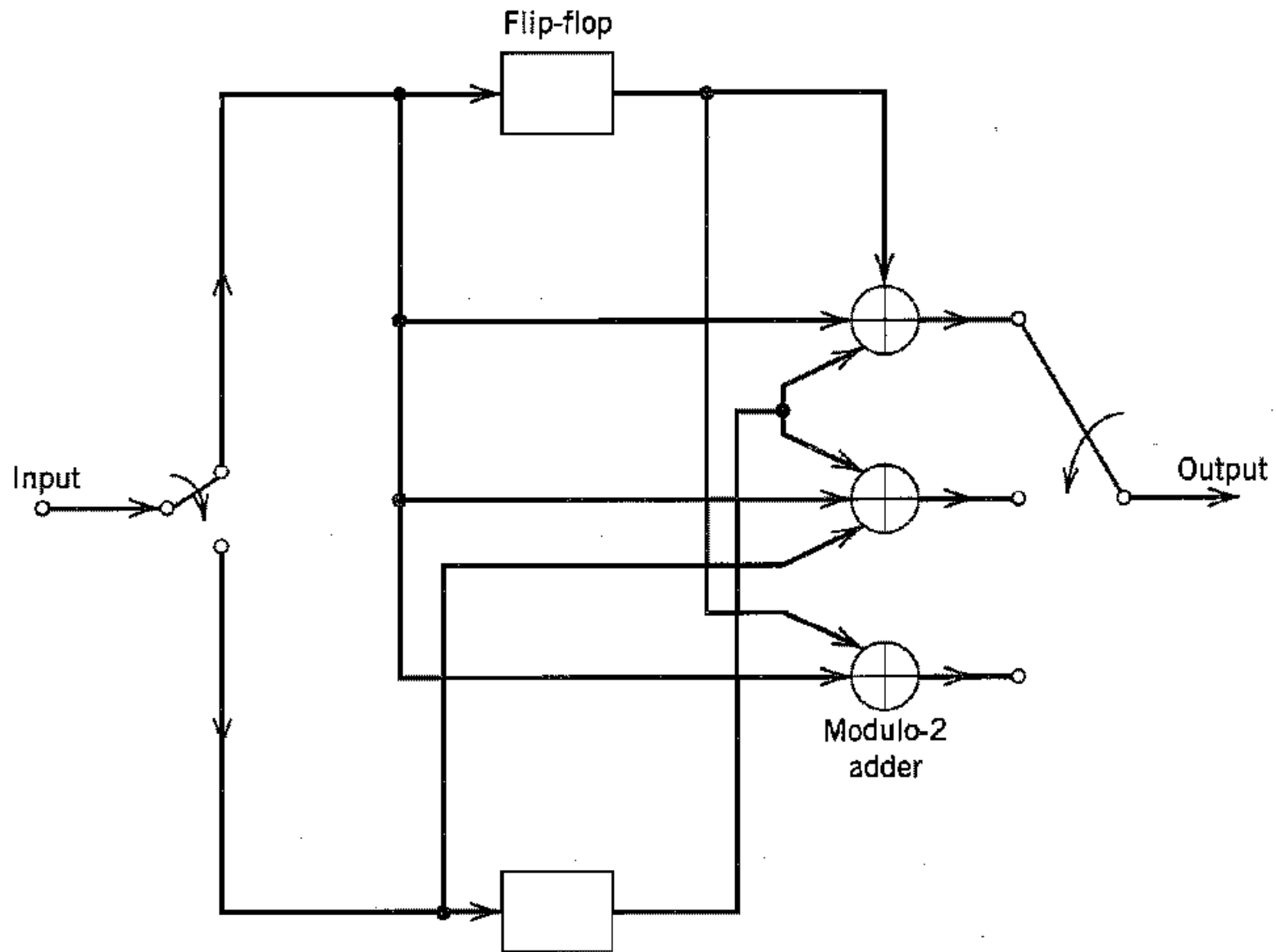
- ▶ Hence the output of path 1 is (1111001) .

- ▶ The output of path 2 is given by

$$c^{(2)}(D) = g^{(2)}(D)m(D) = 1 + D^2 + D^3 + D^4 + D^5 + D^6$$

- ▶ Hence the output of path 2 is (1011111).
- ▶ Multiplexing the two output sequences of path 1 and 2, we get the encoded sequence as $\mathbf{c} = (11, 10, 11, 11, 01, 01, 11)$.

Example



Constraint length-2, rate- $\frac{2}{3}$ encoder

Here we have two inputs ($k = 2$) and three output ($n = 3$). Therefore, there are three generator polynomial corresponding to each input sequence.

We obtain the generator polynomials for the $\frac{2}{3}$ encoder as:

$$g_1^{(1)}(D) = 1 + D; \quad g_1^{(2)}(D) = 1; \quad g_1^{(3)}(D) = 1 + D$$

$$g_2^{(1)}(D) = D; \quad g_2^{(2)}(D) = 1 + D; \quad g_2^{(3)}(D) = 0$$

► Now we can write the encoding equations as

$$c^{(1)}(D) = m_1(D)g_1^{(1)}(D) + m_2(D)g_2^{(1)}(D)$$

$$c^{(2)}(D) = m_1(D)g_1^{(2)}(D) + m_2(D)g_2^{(2)}(D)$$

$$c^{(3)}(D) = m_1(D)g_1^{(3)}(D) + m_2(D)g_2^{(3)}(D)$$

Ex: Let us consider our input as $\mathbf{m} = (11, 01, 10)$, i.e.,

$$m_1(D) = 1 + D^2 \text{ and } m_2(D) = 1 + D$$

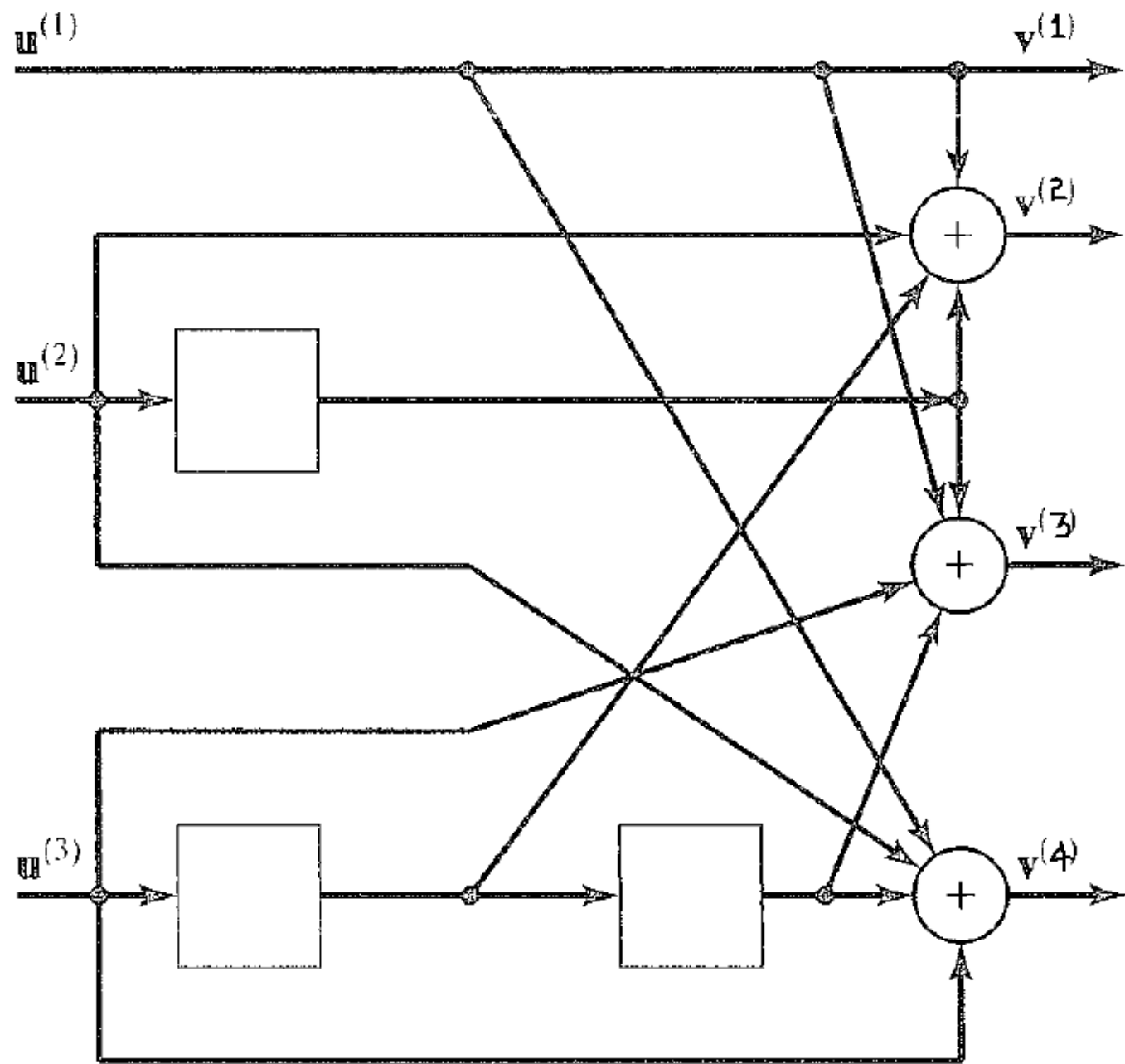
$$\begin{aligned} c^{(1)}(D) &= m_1(D)g_1^{(1)}(D) + m_2(D)g_2^{(1)}(D) \\ &= (1 + D^2)(1 + D) + (1 + D)D \\ &= 1 + D + D^2 + D^3 + D + D^2 = 1 + D^3 \quad (\mathbf{1} \mathbf{0} \mathbf{0} \mathbf{1}) \end{aligned}$$

$$\begin{aligned} c^{(2)}(D) &= m_1(D)g_1^{(2)}(D) + m_2(D)g_2^{(2)}(D) \\ &= (1 + D^2)1 + (1 + D)^2 = 0 \quad (\mathbf{0} \mathbf{0} \mathbf{0} \mathbf{0}) \end{aligned}$$

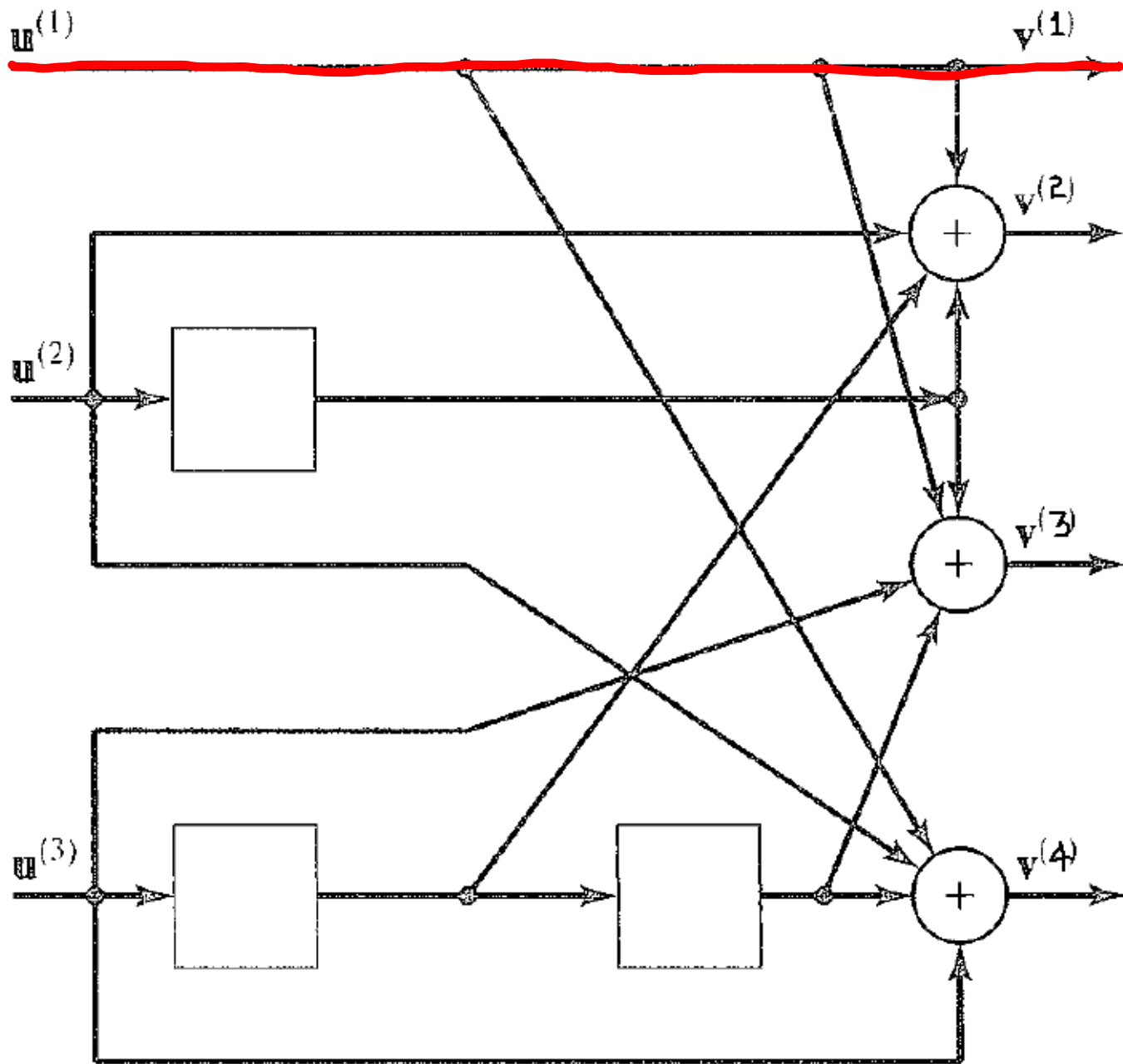
$$\begin{aligned} c^{(3)}(D) &= m_1(D)g_1^{(3)}(D) + m_2(D)g_2^{(3)}(D) \\ &= (1 + D^2)(1 + D) + (1 + D)0 \\ &= 1 + D + D^2 + D^3 \quad (\mathbf{1} \mathbf{1} \mathbf{1} \mathbf{1}) \end{aligned}$$

The overall output is $(\mathbf{1} \mathbf{0} \mathbf{1}, \mathbf{0} \mathbf{0} \mathbf{1}, \mathbf{0} \mathbf{0} \mathbf{1}, \mathbf{1} \mathbf{0} \mathbf{1})$

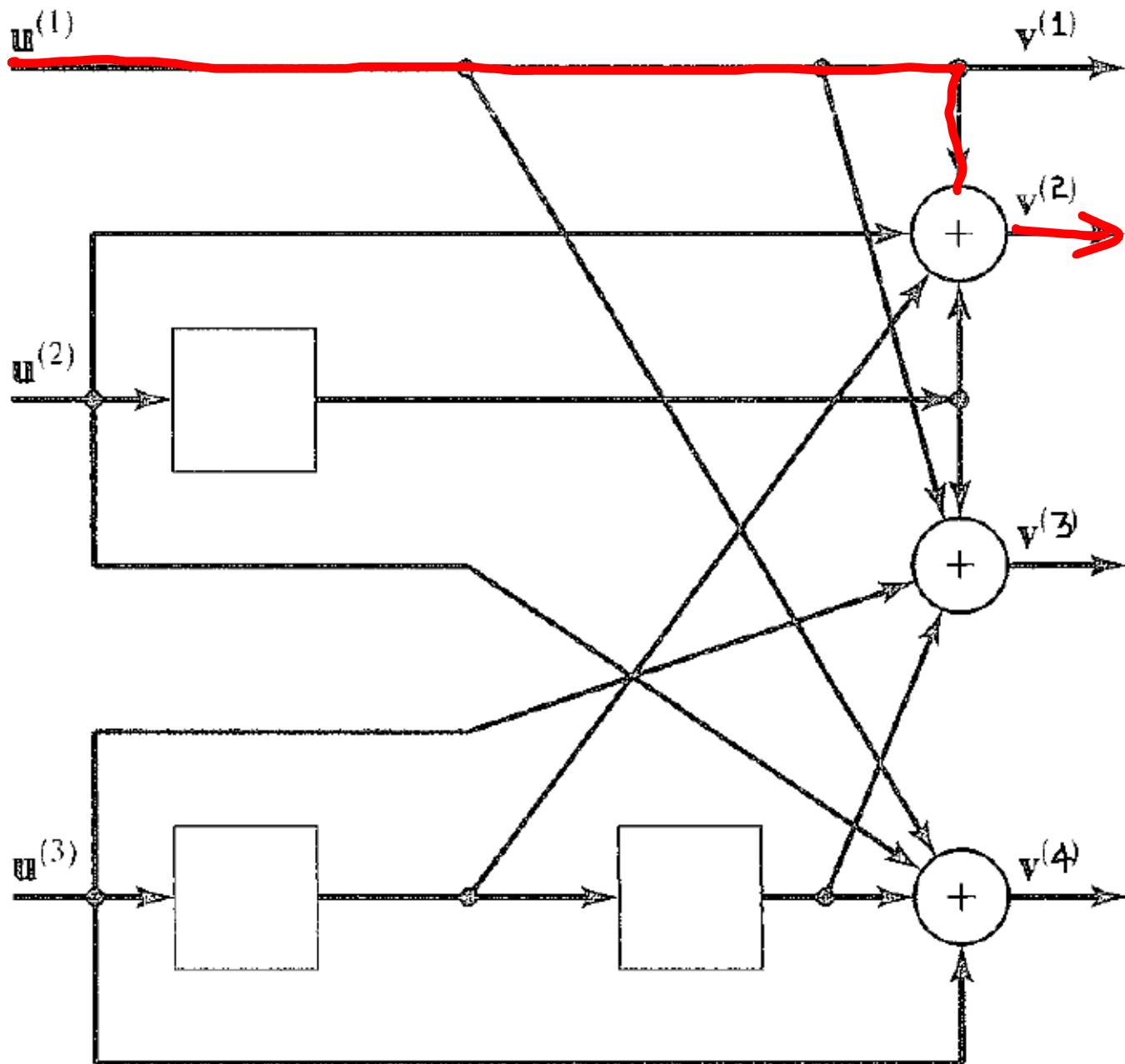
Example



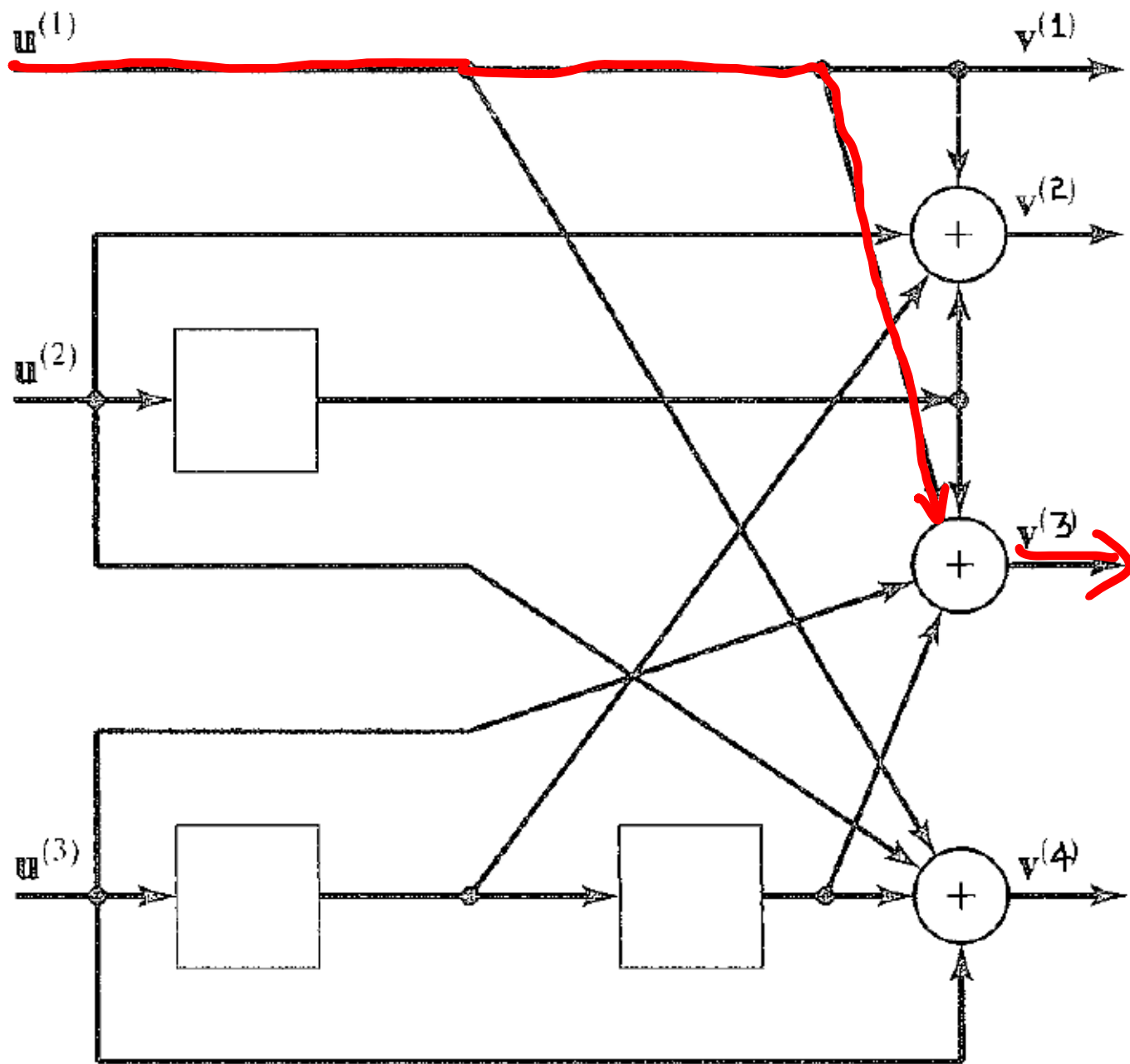
$$g_1^{(1)} = 1$$



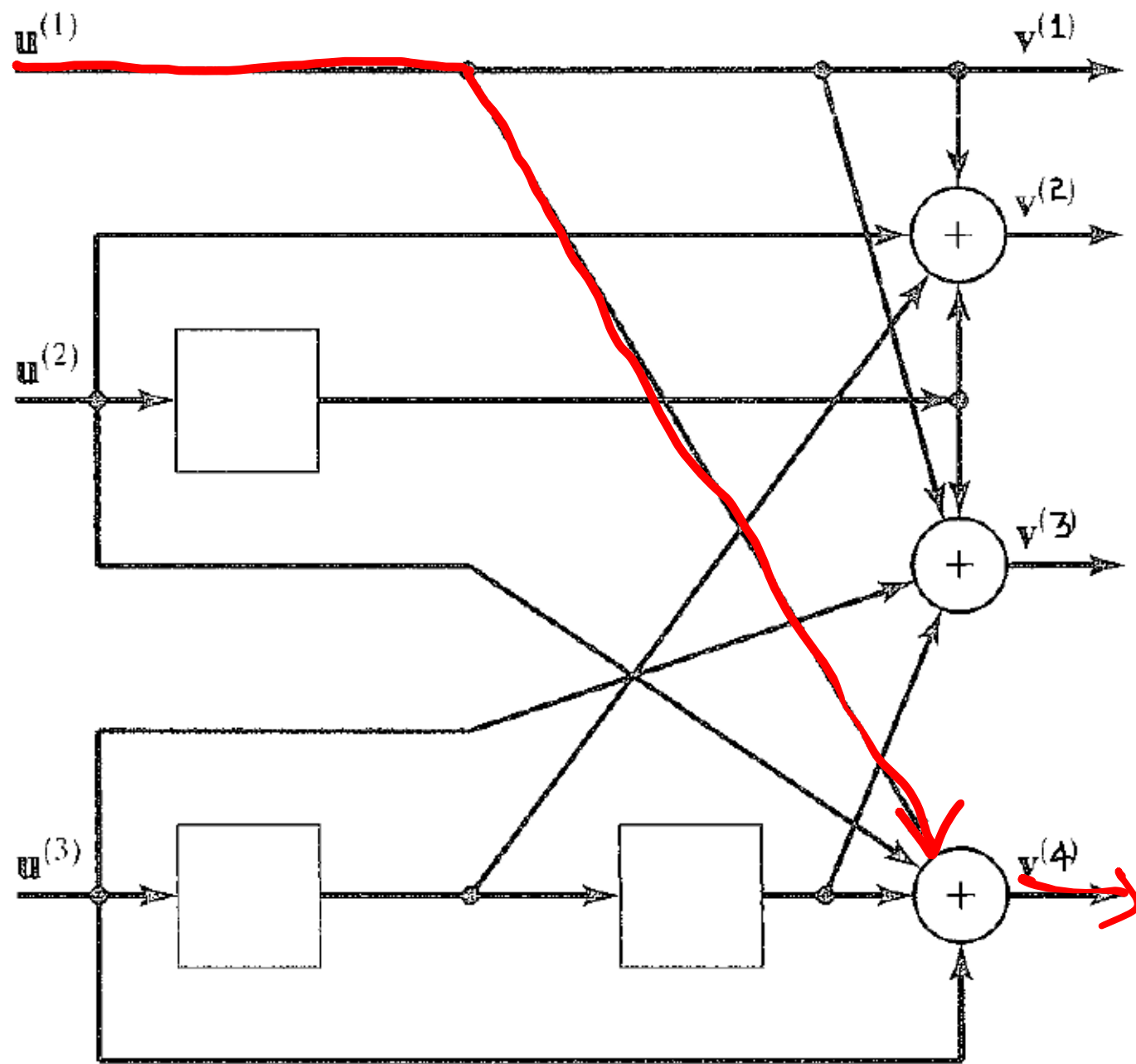
$$g_1^{(2)} = 1$$



$$g_1^{(3)} = 1$$

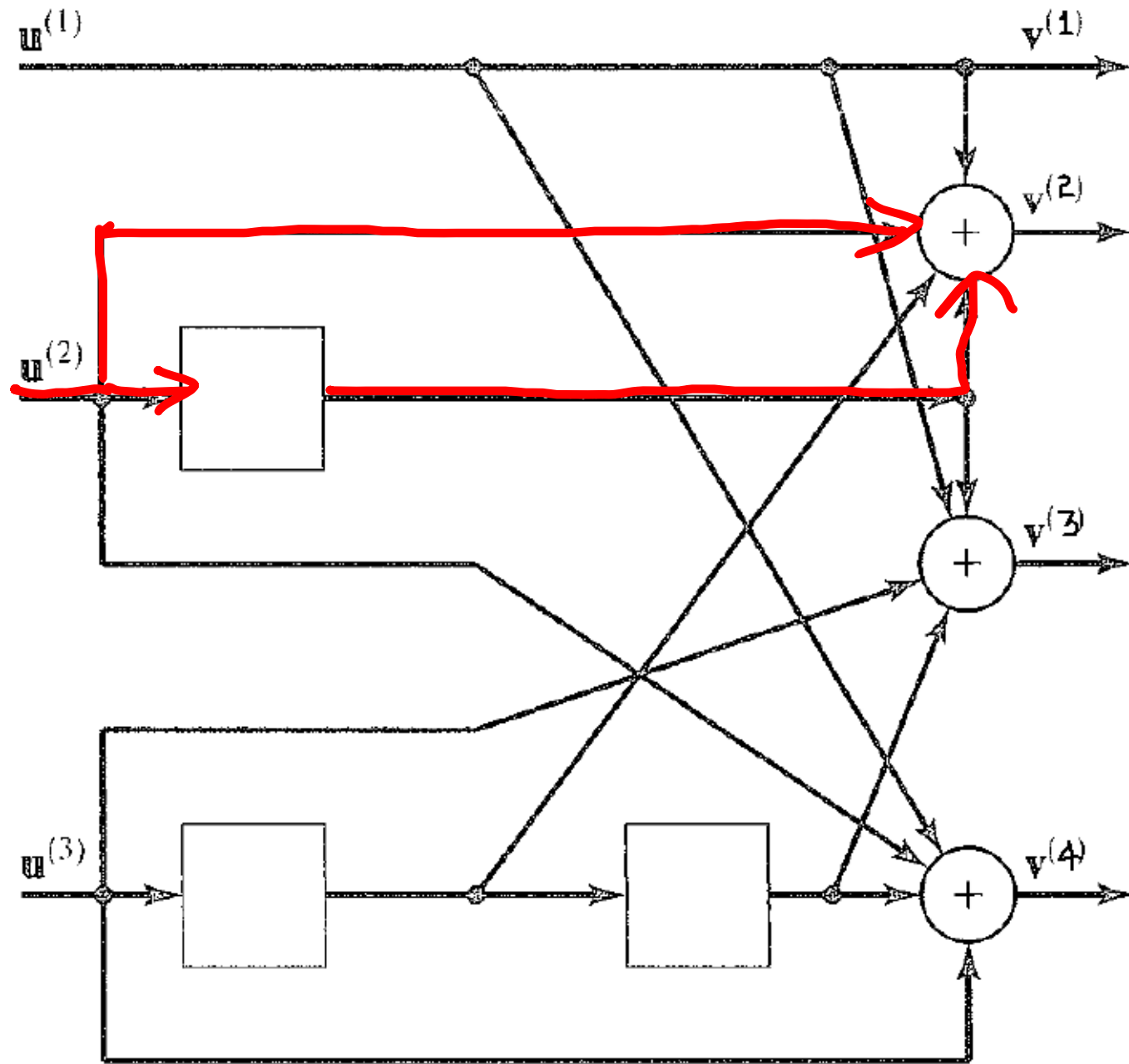


$$g_1^{(4)} = 1$$



$$g^{(1)}_2 = 0$$

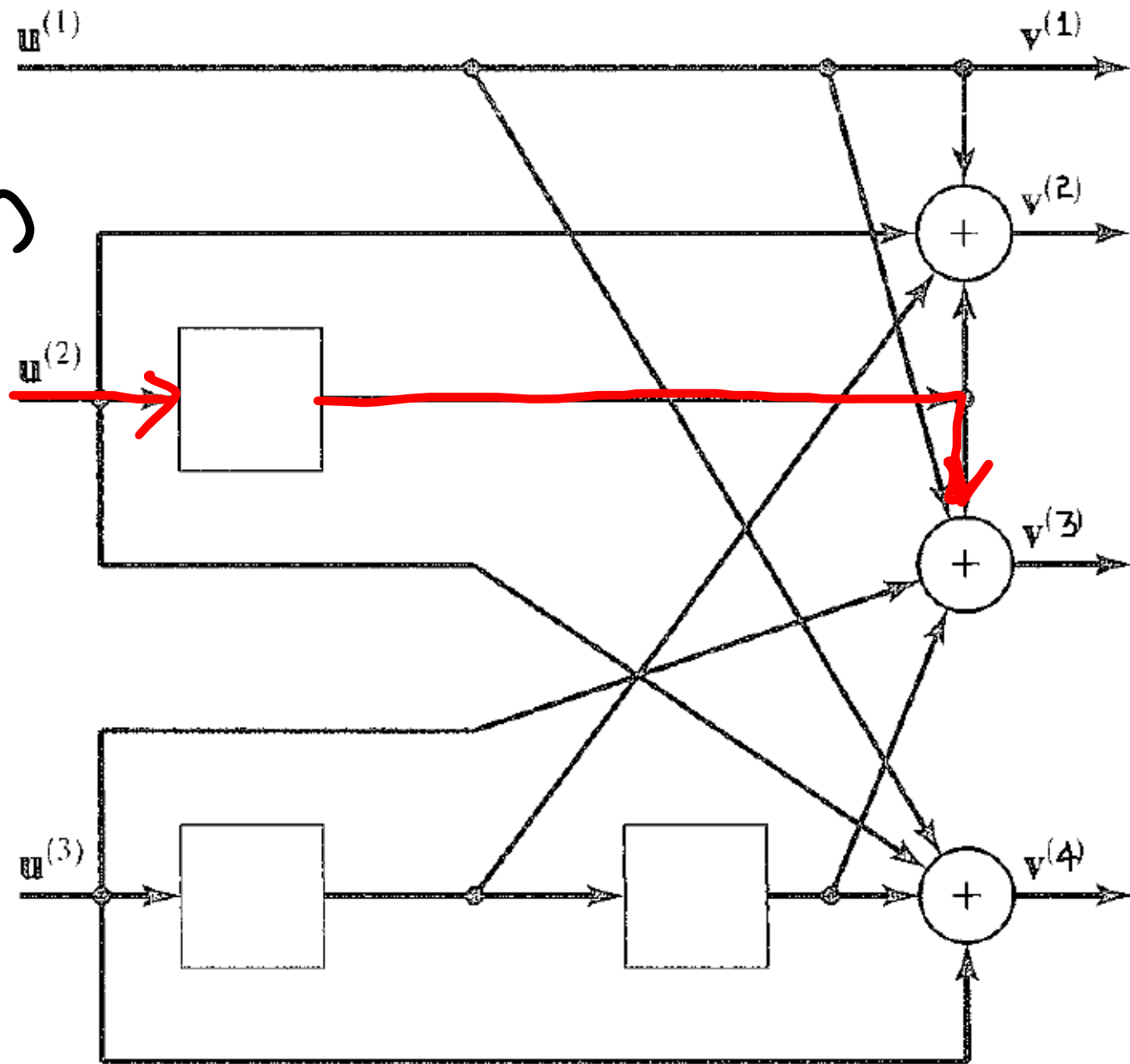
$$g^{(k)}_2 = (110) = 110$$



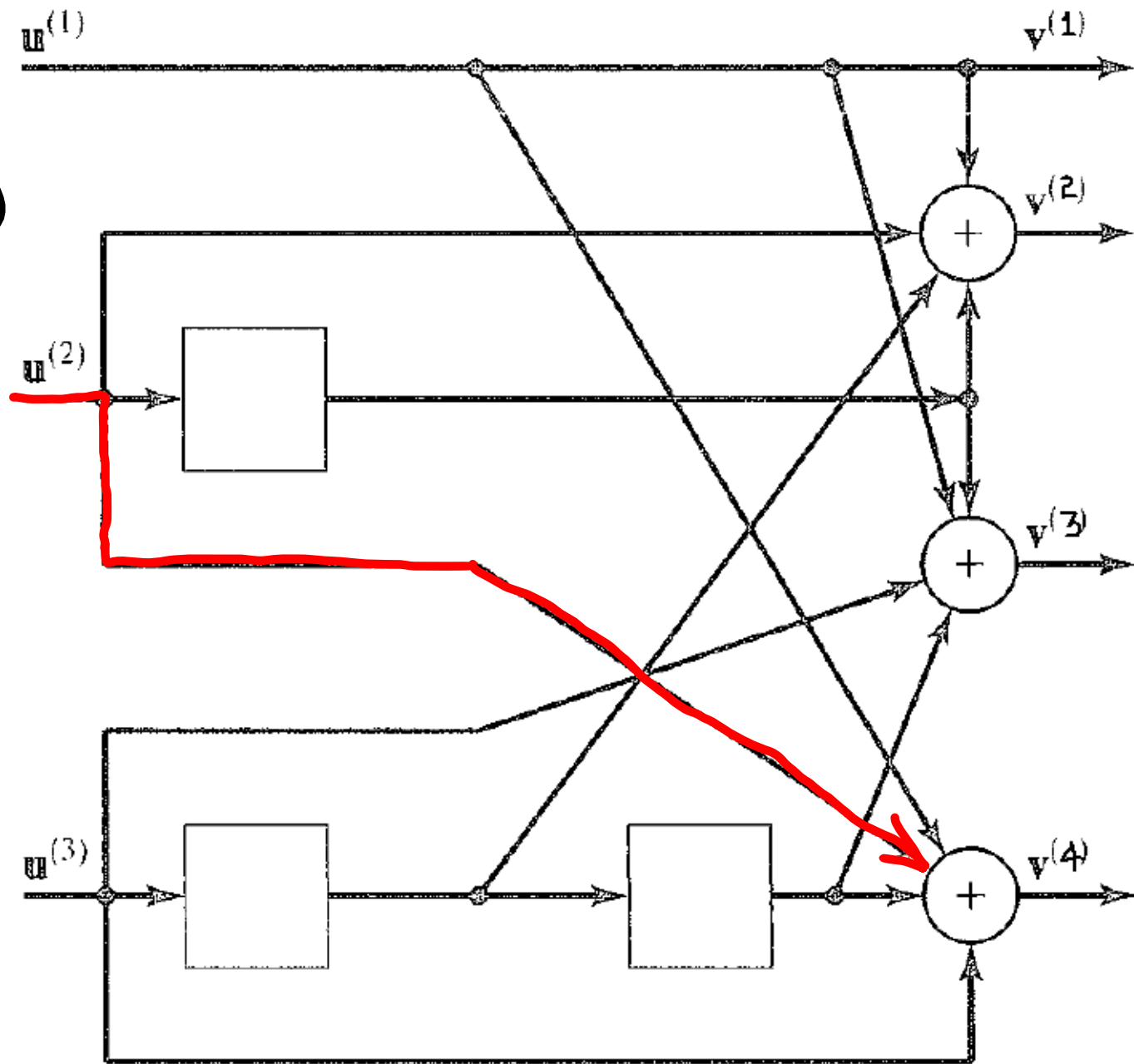
$$g(3)$$

$$v_2 = (010)$$

$$= D$$

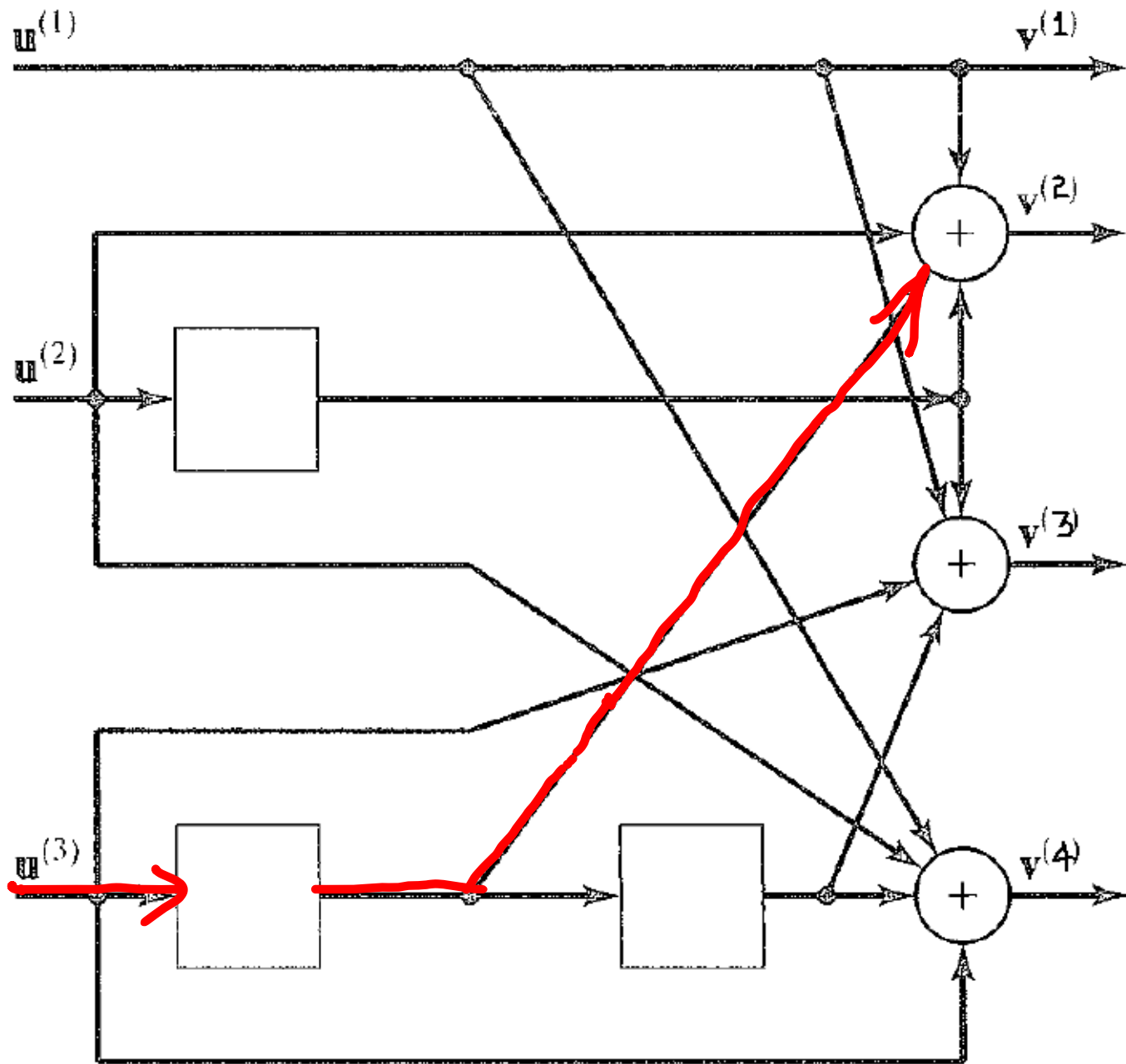


$$g_2^{(4)} = (100)$$



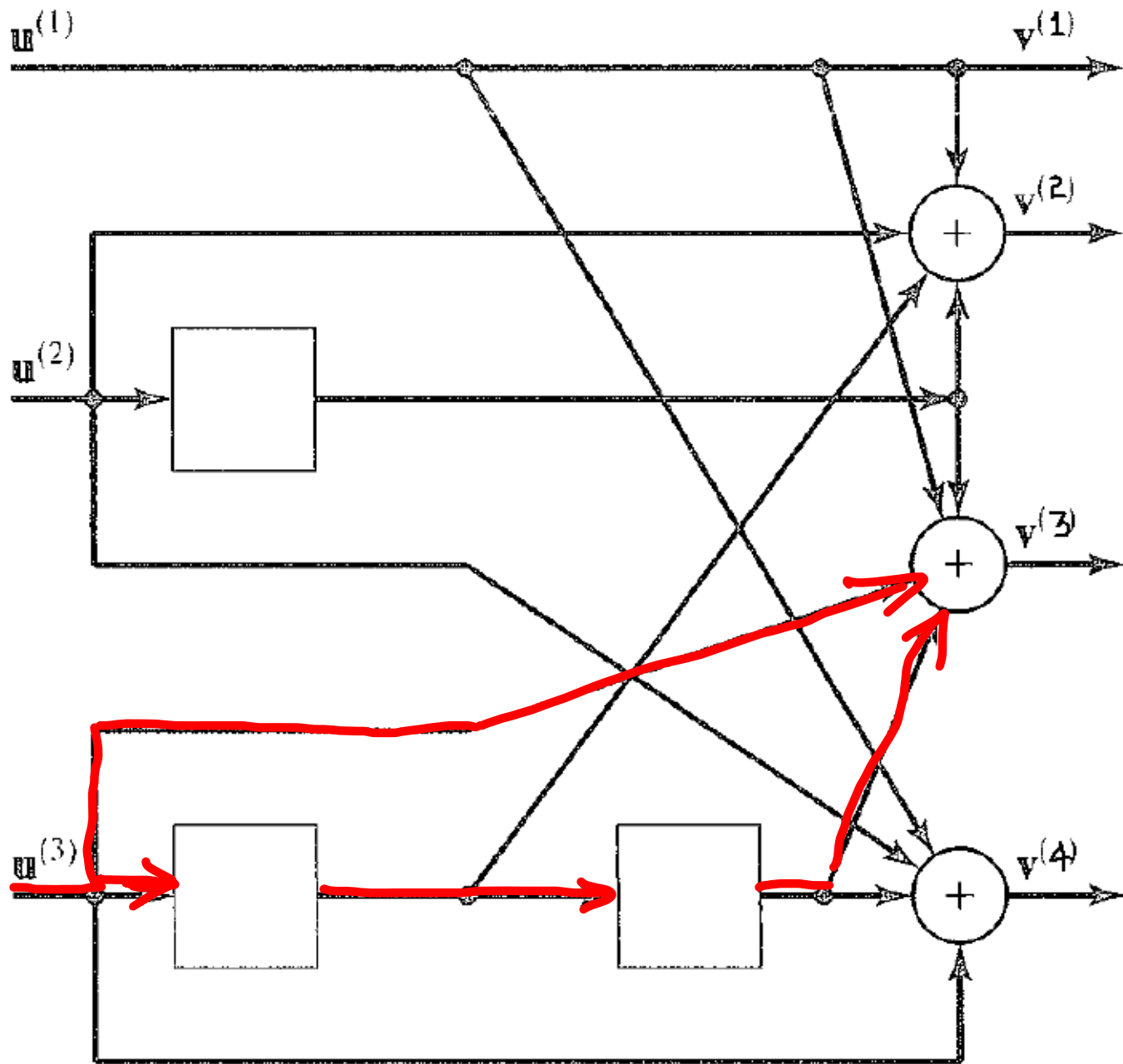
$$g_3^{(1)} = 0$$

$$g_3^{(2)} = (0 \ 1 \ 0) \\ = D$$



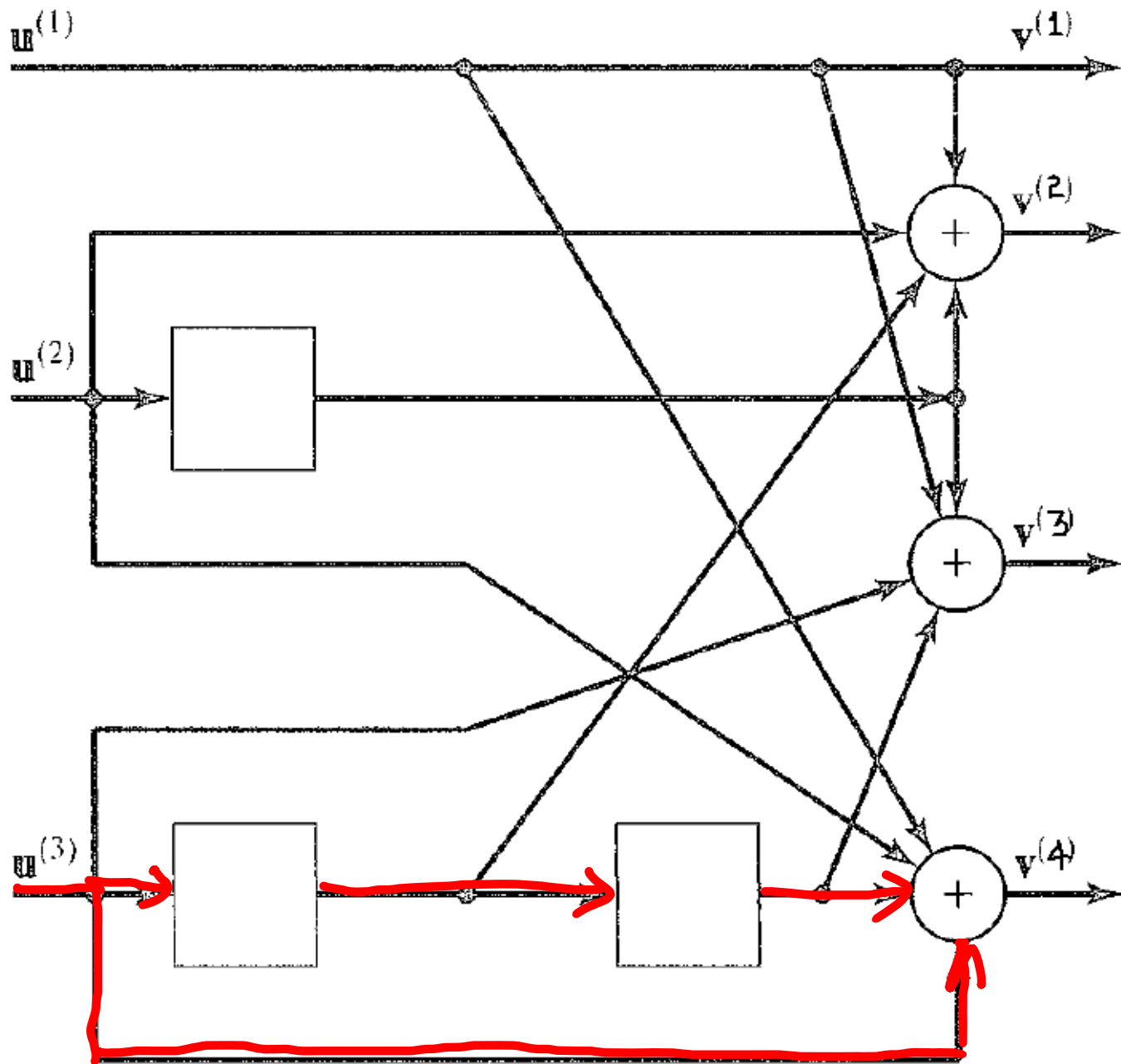
$$g^{(3)}_3 = (1 \ 0 \ 1)$$

$$= 1 + D^2$$



$$g_3^{(4)} = (101)$$

$$= 1 + D^2$$



The generator sequences are:

$$g_1^{(1)} = (1\ 0\ 0) \Rightarrow 1; \quad g_1^{(2)} = (1\ 0\ 0) \Rightarrow 1;$$
$$g_1^{(3)} = (1\ 0\ 0) \Rightarrow 1; \quad g_1^{(4)} = (1\ 0\ 0) \Rightarrow 1$$

$$g_2^{(1)} = (0\ 0\ 0) \Rightarrow 0; \quad g_2^{(2)} = (1\ 1\ 0) \Rightarrow 1 + D;$$
$$g_2^{(3)} = (0\ 1\ 0) \Rightarrow D; \quad g_2^{(4)} = (1\ 0\ 0) \Rightarrow 1$$

$$g_3^{(1)} = (0\ 0\ 0) \Rightarrow 0; \quad g_3^{(2)} = (0\ 1\ 0) \Rightarrow D;$$
$$g_3^{(3)} = (1\ 0\ 1) \Rightarrow 1 + D^2; \quad g_3^{(4)} = (1\ 0\ 1) \Rightarrow 1 + D^2$$

Ex: Let us consider our input as $u = (111, 011, 100, 101)$, i.e.,

$$u_1 = (1\ 0\ 1\ 1) \Rightarrow u_1(D) = 1 + D^2 + D^3$$

$$u_2 = (1\ 1\ 0\ 0) \Rightarrow u_2(D) = 1 + D$$

$$u_3 = (1\ 1\ 0\ 1) \Rightarrow u_3(D) = 1 + D + D^3$$

The corresponding out sequences are obtained as:

$$v^{(1)}(D) = v_1^{(1)}(D) + v_2^{(1)}(D) + v_3^{(1)}(D)$$

$$v^{(1)}(D) = u_1(D) g_1^{(1)}(D) + u_2(D) g_2^{(1)}(D) + u_3(D) g_3^{(1)}(D)$$

$$v^{(1)}(D) = (1 + D^2 + D^3) \times 1 + (1 + D) \times 0 + (1 + D + D^3) \times 0$$

$$v^{(1)}(D) = 1 + D^2 + D^3$$

$$v^{(2)}(D) = v_1^{(2)}(D) + v_2^{(2)}(D) + v_3^{(2)}(D)$$

$$v^{(2)}(D) = u_1(D) g_1^{(2)}(D) + u_2(D) g_2^{(2)}(D) + u_3(D) g_3^{(2)}(D)$$

$$v^{(2)}(D) = (1 + D^2 + D^3) \times 1 + (1 + D) \times (1 + D) + (1 + D + D^3) \times D$$

$$v^{(2)}(D) = D + D^2 + D^3 + D^4$$

$$v^{(3)}(D) = v_1^{(3)}(D) + v_2^{(3)}(D) + v_3^{(3)}(D)$$

$$v^{(3)}(D) = u_1(D) g_1^{(3)}(D) + u_2(D) g_2^{(3)}(D) + u_3(D) g_3^{(3)}(D)$$

$$v^{(3)}(D) = (1 + D^2 + D^3) \times 1 + (1 + D) \times D + (1 + D + D^3) \times (1 + D^2)$$

$$v^{(3)}(D) = D^2 + D^3 + D^5$$

$$v^{(4)}(D) = v_1^{(4)}(D) + v_2^{(4)}(D) + v_3^{(4)}(D)$$

$$v^{(4)}(D) = u_1(D) g_1^{(4)}(D) + u_2(D) g_2^{(4)}(D) + u_3(D) g_3^{(4)}(D)$$

$$v^{(4)}(D) = (1 + D^2 + D^3) \times 1 + (1 + D) \times 1 + (1 + D + D^3) \times (1 + D^2)$$

$$v^{(4)}(D) = 1 + D^3 + D^5$$

The individual outputs are:

$$v^{(1)} = (1 \ 0 \ 1 \ 1 \ 0 \ 0)$$

$$v^{(2)} = (0 \ 1 \ 1 \ 1 \ 1 \ 0)$$

$$v^{(3)} = (0 \ 0 \ 1 \ 1 \ 0 \ 1)$$

$$v^{(4)} = (1 \ 0 \ 0 \ 1 \ 0 \ 1)$$

The overall output is (1 0 0 1, 0 1 0 0, 1 1 1 0, 1 1 1 1, 0 1 0 0, 0 0 1 1)